

Session 3 Documentation on SPSS work

Topic 1: Creating Variables in SPSS

Learn how to create new variables in SPSS using **Variable View** and **Data View**.

Instructions

Step 1 – Open the Variable View

1. Open your SPSS dataset or create a new file (**File → New → Data**).
2. At the bottom left of the SPSS window, click on **Variable View**.
 - You'll see columns like *Name, Type, Width, Decimals, Label, Values, Missing, Columns, Align, Measure, and Role*.

Step 2 – Define the Variable

In the first empty row, start entering the details of your new variable:

Column	What It Means	Example Entry
Name	The variable's short name (no spaces, do not start with a number or character)	Age, Gender, Salary
Type	Defines data type	Click cell → choose Numeric, String, Date , etc.
Width	Maximum number of digits or characters	Default is fine for most cases
Decimals	Number of decimals displayed	Use 0 for whole numbers like Age
Label	Descriptive name (can include spaces and numbers)	Participant Age in Years
Values	Used for categorical variables to assign labels	Example: 1 = Male, 2 = Female
Missing	Specify values for missing data	
Measure	Defines the measurement scale	Choose Nominal, Ordinal, or Scale

Step 3 – Switch to Data View

1. Click on the **Data View** tab at the bottom.
2. Each column corresponds to a variable you defined.

3. Enter data manually or paste it from Excel.

Step 4 – Save Your Dataset: Click **File → Save As...**

Topic 2: Dealing with and Deleting Duplicates in SPSS

Learn how to **identify** and **remove duplicate cases** (rows) in SPSS to clean your dataset before analysis.

Instructions

Step 1 – Identify Duplicates

SPSS does not automatically highlight duplicates, but you can flag them using the **Identify Duplicate Cases** function.

1. Go to **Data → Identify Duplicate Cases...**
2. Move the variable(s) you want to check for duplicates to the box labeled **Define matching cases by**:
 - Example: move ID if you want to check for duplicate IDs.(it is how you define duplicates and based on what variables depending on the dataset)
3. Under **Sort within matching groups by**, select a variable that indicates order (e.g., Date or Case Number) if relevant.
4. In the section **Create indicator variable**, give a name for the new variable (e.g., Dup_Flag).
5. Click **OK**.

Step 2 – Review the New Variable

- SPSS will create a new variable (PrimaryLast or PrimaryFrist, or whatever name you choose) that indicates whether a case is a duplicate:
 - Primary → the first occurrence of each unique record.
 - Duplicate → all subsequent identical records.

Check your **Data View** to see this new column.

Step 3 – Filter or Delete Duplicate Cases

Option 1 – Delete duplicates directly

1. Go to **Data → Select Cases...**
2. Choose **If condition is satisfied**, then click **If...**
3. Type this condition:
4. `PrimaryFirst = 1`
5. Check: delete unselected cases.

Option 2 – Keep only unique cases “filter”

1. In the **Select Cases** dialog box, choose:
 - **Filter out unselected cases.**
2. Click **OK**.
 - Only the first instance (Primary) will remain active for analysis.

Step 4 – Save the Cleaned File

1. Save your cleaned dataset using **File → Save As...**

Topic 3: Dealing with Inconsistencies in SPSS

Learn how to **detect and fix inconsistent data entries** (e.g., “USA” vs “United States,” “male” vs “Male”) to ensure your dataset is clean and standardized before analysis.

Instructions

Step 1 – Understand What Inconsistencies Are

Inconsistencies occur when the same category is entered in different ways.
Examples:

Inconsistent Entries	Correct Version
USA, U.S.A, United States, united states of america	United States
male, Male, M	Male
NY, N.Y., New York	New York

These differences cause SPSS to treat each as a **separate category**, which can distort frequency counts and statistical results.

Step 2 – Explore the Data

1. Go to **Analyze → Descriptive Statistics → Frequencies...**
2. Move the variable you suspect has inconsistencies (e.g., Gender) to the **Variable(s)** box.
3. Click **OK**.
4. Check the **Output Window** for multiple similar categories — these reveal inconsistencies.

Step 3 – Correct Inconsistencies Using Recode Function

Option 1 – Recode into Same Variables (Overwrite Existing Data)

1. Go to **Transform → Recode into Same Variables...**
2. Move the inconsistent variable (e.g., Gender) to the **Numeric or String Variable → Output Variable** box.
3. Click **Old and New Values...**
4. Under **Old Value**, type one inconsistent version (e.g., usa), then under **New Value**, type the corrected one (United States).
5. Click **Add**.
6. Repeat this for all inconsistent forms (e.g., U.S.A, united states of america).
7. Click **Continue → OK**.
8. SPSS replaces all inconsistent entries with your chosen standardized label.

Option 2 – Recode into Different Variable (Keep Original Data Safe)

1. Go to **Transform → Recode into Different Variables...**
2. Move the variable (e.g., Gender) into the **Input Variable → Output Variable** box.
3. Name the new variable (e.g., Gender_Clean) and click **Change**.
4. Click **Old and New Value**.
5. Enter each inconsistent form under *Old Value* and the standardized form under *New Value*.
6. Click **Add** for each mapping, then **Continue → OK**.
7. A new cleaned variable will appear in **Data View**.

Step 4 – Use Transformations to Fix Text Case

If inconsistencies are caused by letter case (e.g., “male” vs “Male”):

1. Go to **Transform → Compute Variable...**
2. In **Target Variable**, type a new name (e.g., Gender_Upper).
3. In **Numeric Expression**, type one of the following functions:
 - UPCASE(Gender) → converts all letters to uppercase.
 - LOWER(Gender) → converts all letters to lowercase.
 - PROPER(Gender) → capitalizes the first letter of each word.
4. Click **OK**.
5. The new standardized text variable will appear in your **Data View**.

Step 5 – Verify the Corrections

1. Run **Analyze → Descriptive Statistics → Frequencies...** again on your new variable (e.g., Gender_Clean or Gender_Upper).
2. Confirm that inconsistent categories no longer appear.

Topic 4: Handling Missing Values in SPSS

Learn how to **detect**, **label**, **exclude**, or **replace** missing values in your dataset to ensure your analysis is accurate and complete.

Instructions

Step 1 – Understand What Missing Values Are

Missing values are empty cells or special codes (e.g., 99, 999, NA) that indicate unavailable data. SPSS treats empty cells as *system-missing* and user-defined codes (like 99) as *user-missing* when specified.

Example:

ID Age Income

1	25	3500
2		4200
3	30	999

Here, case 2 has a **system-missing** value for Age, and case 3 may have **user-missing** income (coded as 999).

Step 2 – Identify Missing Values

1. Go to **Analyze → Descriptive Statistics → Frequencies...**
2. Move the variable(s) (e.g., Age, Income) to the **Variable(s)** box.
3. Click **OK**.
4. In the output, check:
 - The number of **Valid** vs **Missing** cases.
 - Unusual codes (e.g., 99 or 999) that should represent missing data.

Step 3 – Define User-Missing Values

If you use special codes like 99, you need to tell SPSS to treat them as missing.

1. Switch to **Variable View**.
2. Locate the variable (e.g., Income).
3. Click the cell under **Missing**.
4. Choose **Discrete missing values**, then enter up to three codes (e.g., 99, 999).
5. Click **OK**.
 - SPSS now treats these as missing during analysis.

Step 4 – Decide How to Handle Missing Values

Option 1 – Exclude Missing Values from Analysis

1. In any analysis dialog (e.g., **Analyze → Correlate → Bivariate**), look for the “**Missing Values**” option.
2. Choose **Exclude cases listwise** or **pairwise** as needed.

Option 2 – Delete Missing Rows Permanently

If you want to permanently remove rows with missing data:

1. Go to **Data → Select Cases...**

2. Choose **If condition is satisfied** → click **If...**
3. Type a condition like:
4. NOT MISSING(Age)

(This keeps only rows where Age is not missing.)

5. Click **Continue** → **OK**.
6. SPSS will mark excluded rows with a slash; go to **Data** → **Copy** to keep only the filtered dataset.

Option 3 – Replace Missing Values (Imputation)

If you want to estimate missing values instead of deleting them:

A. Using Replace Missing Values tool:

1. Go to **Transform** → **Replace Missing Values...**
2. Move the variable(s) with missing data to the **New Variable(s)** box.
3. Under **Method**, choose how SPSS should fill the gaps:
 - **Series Mean:** replaces with the variable's mean.
 - **Median of nearby points:** for ordered data.
 - **Linear Interpolation:** estimates based on surrounding values (for time-series data).
 - **Linear Trend at Point:** predicts based on trend line.
4. Click **OK**.
5. SPSS creates new variables ending in _1 (e.g., Income_1) containing the replaced values.

Step 5 – Verify Your Results

1. Check the new variable in **Data View** — missing values should be replaced or labeled.
2. Run **Analyze** → **Descriptive Statistics** → **Frequencies** again to confirm no missing values remain.